西南财经大学本科期末考试试题册（A）

（2020－2021学年第1学期）

以下各项由命题教师填写：

课程名称：面向对象程序设计（JAVA SE）(英)

命题教师：王磊

适用对象（年级专业）**：**2020 信息管理与信息系统专业

使用试题的任课教师姓名：王磊

试题说明：

1、考试类型：闭卷[ ]　开卷[√]

2、本套试题共 **2** 道大题，共 **7** 页，完卷时间 120 分钟。

3、考试用品中除纸、笔、尺子外，可另带的用具有：

计算器[ ]　　字典[ ]　等

（请在下划线上填上具体数字或内容，所选[　　]内打钩）

考试时间（由制卷方填写）**：**

以下各项由学生填写：

任课教师：　　　　　　　　　　　**年级专业：**

学生姓名：　　　　　　　　　　　**学　　号：**

考生注意事项：1.出示学生证（或身份证）和准考证于桌面左上角，以备查验。

2.拿到试卷后清点并检查试卷页数，如有重页、页数不足、空白页及印刷模糊等举手向监考教师示意调换试卷。

3.答题前先将试题册及答题纸上的任课教师、专业、年级、学号、姓名填写完整。

4.所有答案均需填写在答题纸上，答在试题册上无效。

5.考生不得携带任何通讯工具进入考场。

6.考试结束后，将试题册、答题纸和草稿纸等下发材料上交监考教师，不得带离考场。

7.严格遵守考场纪律。

Part I: Single Choice Questions:

Mark your answers on below sheet (total 30 pts, 3 pts each)

Answer sheet:

1	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**

1. If you attempt to add an **int**, a **byte**, a **long**, and a **double**, the result will be a ______ value.

A. int

B. long

C. byte

D. double

2. The following code displays ____________.

```
double temperature = 50;
if (temperature >= 100)
      System.out.println("too hot");
else if (temperature <= 40)
      System.out.println("too cold");
else
      System.out.println("just right");
```

A. too hot

B. too cold

C. too hot too cold just right

D. just right

3. How many times will the following code print "Welcome to Java"?____________。

```
int count = 0;
```

```
do {
      System.out.println("Welcome to Java");
} while (count++ < 10);
```

A. 10

B. 8

C. 11

D. 9

4. What is the output for y? __________.

```
int y = 0;
for (int i = 0; i<10; ++i) {
      y += i;
}
System.out.println(y);
```

A. 45

B. 12

C. 13

D. 10

5. Analyze the following code: __________.

```
public class Test {
      public static void main(String[] args) {
            System.out.println(xMethod(5, 500L));
      }
      public static int xMethod(int n, long l) {
            System.out.println("int, long");
            return n;
      }
      public static long xMethod(long n, long l) {
            System.out.println("long, long");
```

```
            return n;
        }
    }
```

A. The program does not compile because the compiler cannot distinguish which xmethod to invoke.

B. The program runs fine but displays things other than 5.

C. The program displays long, long followed by 5.

D. The program displays int, long followed by 5.

6. How many elements are in array double[] list = new double[5]? ___________

A. 5

B. 4

C. 6

D. 0

7. Given the declaration Circle x = new Circle( ), which of the following statement is most accurate. ___________

A. x contains an object of the Circle type.

B. You can assign an int value to x.

C. x contains a reference to a Circle object.

D. x contains an int value.

8. What is the value of myCount.dCount displayed? ___________

```
  public class Test {
      public static void main(String[] args) {
          Count myCount = new Count();
          int times = 0;

          for (int i=0; i < 10; i++)
              increment(myCount, times);
          System.out.println(
```

```
                "myCount.count =      " + myCount.dCount);
            System.out.println("times = "+ times);
        }
        public static void increment(Count c, int times) {
            c.dCount++;
            times++;
        }
    }

    class Count {
        int dCount;
        Count(int c) {
            dCount = c;
        }
        Count( ) {
            dCount = 1;
        }
    }
```

A. 9

B. 10

C. 11

D. 0

9. A subclass inherits ______________ from its superclass.

A. private method

B. protected method

C. public method

D. B and C

10. Show the output of running the class Test in the following code lines: _____________

```
interface A {    void print();    }
```

```
class C { }

class B extends C implements A {
    public void print() { }
}

class Test {
    public static void main(String[] args) {
        B b = new B();
        if (b instanceof A)
            System.out.println("b is an instance of A");
        if (b instanceof C)
            System.out.println("b is an instance of C");
    }
}
```

A. Nothing.

B. b is an instance of A.

C. b is an instance of C.

D. b is an instance of A followed by b is an instance of C.

Part II: Write Programs.(total 70 pts)

1. (15 pts) Write a program that prompts the user to enter an integer **n** (assume n >= 3) and displays its largest factor other than itself. (Note: factor is the number that n can be exact devided by it)

2. (15 pts) Write the following method that returns **true** if the list is already sorted in increasing order.

public static boolean isSorted(int[ ] list)

Write a test program that prompts the user to enter a number list and displays whether the list is sorted or not. Here is a sample run. Note that the first number in the input list indicates the number of the elements in the list.

[Sample run 1]:

Please enter number list: 8 14 1 15 16 62 9 12 1

<Output> The list is not sorted

[Sample run 2]:

Please enter number list: 10 1 2 3 4 6 7 7 9 14 23

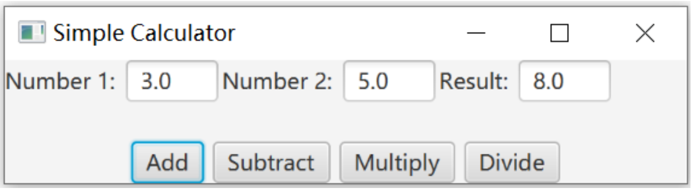
<Output> The list is already sorted

3. (20 pts) Design a class named **Person** and its two subclasses named **Student** and **Staff**.

- A person object has a name, an address, a phone number, and an email address.
- A student object has a status (such as "freshman", "sophomore", "junior", or "senior").
- A staff object has a department (such as "finance", "engineering", "market") and a salary.
- The **Person** class has a **showInfo( )** method to display information of the object, the subclasses **Student** and **Staff** should override the **showInfo( )** method.

Write a test program: create a **Student** and a **Staff** object respectively in its **main( )** method, and display information of them.

4. (20 pts) Write a program to perform addition, subtraction, multiplication and division, as shown in below figure.



Requirements:

- Use suitable Java FX GUI components.
- Use suitable layout to organize components.
- At least contains one event handling process.